

Smart Campus VMS — Data Flow Diagram

Data Flow Diagram (DFD) Documentation

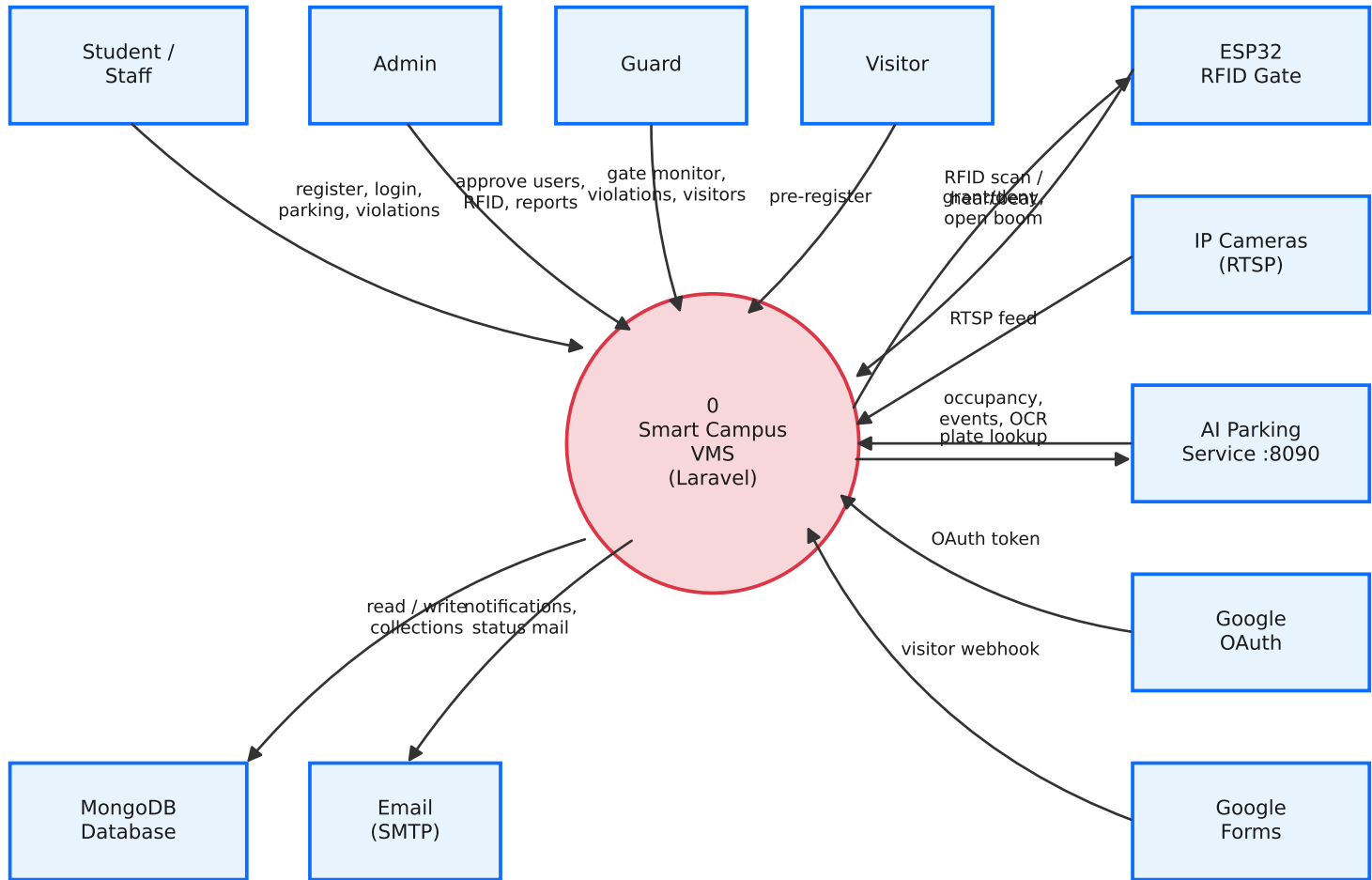
Notation



Smart Campus Vehicle Management System (VMS)
Laravel 12 + MongoDB | ESP32 RFID Gates | YOLO AI Parking | Google OAuth/Forms

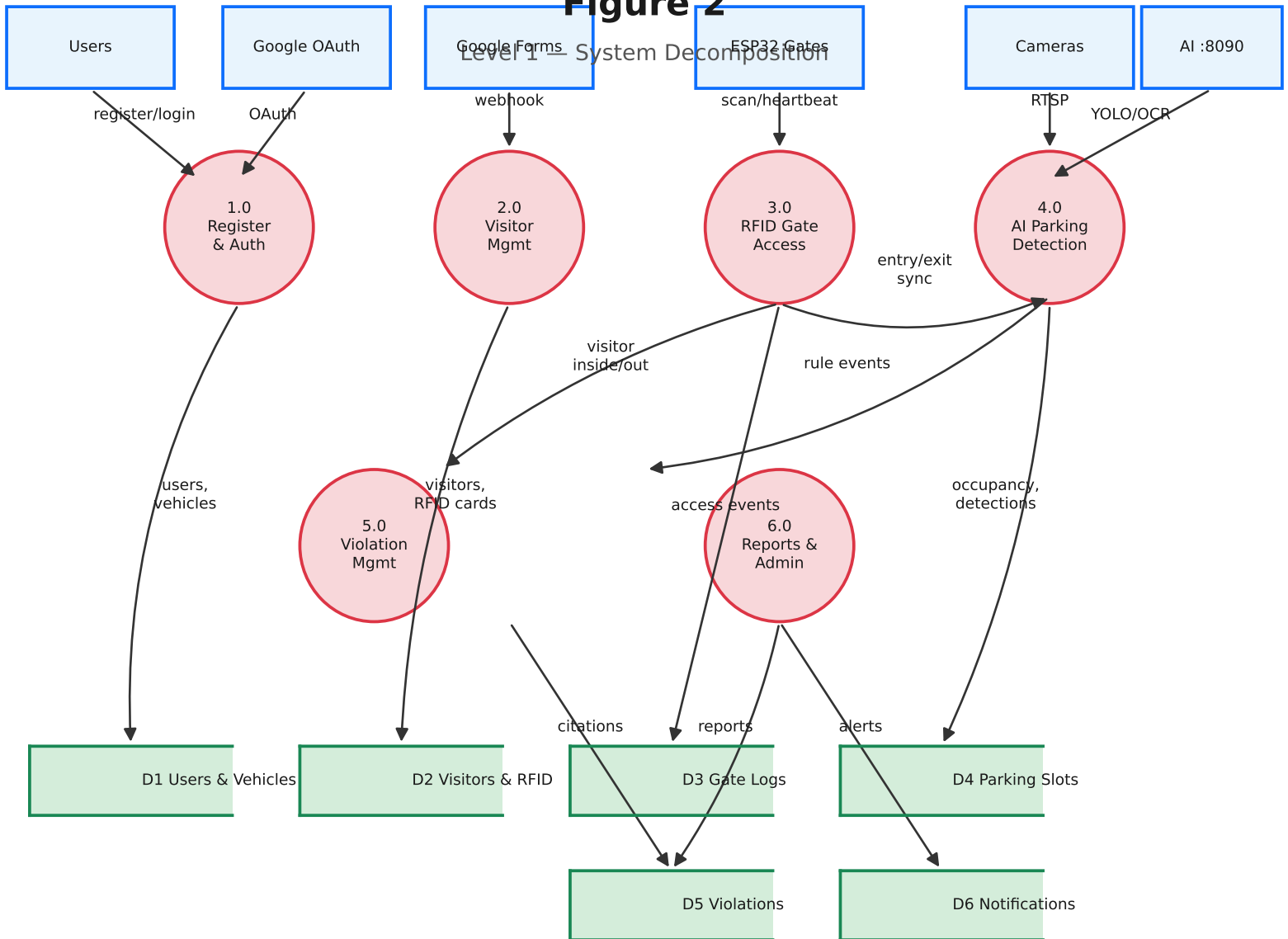
Figure 1

Level 0 — Context Diagram



Level 0 — Context Diagram: Laravel is the central hub for all web, hardware, and AI traffic.

Figure 2



Level 1 — Decomposition: six core processes and six MongoDB-backed data stores.

Figure 3

Level 1 — Process Specifications

Level 1 — Process Details & API Endpoints

1.0 Registration & Authentication

POST /register, /register/scan-id (Campus ID OCR) → D1 users, user_vehicles
POST /login, GET /auth/google → session; admin approves via /admin/registrations

2.0 Visitor Management

POST /visitor/pre-register or /api/visitor/pre-register/google → D2 visitors
Guard assigns RFID: POST /guard/visitors/{id}/assign-rfid → visitor_rfid_cards

3.0 RFID Gate Access

ESP32 POST /api/rfid/scan (X-RFID-TOKEN) → RfidAccessService → D3 gate_logs
POST /api/rfid/heartbeat → boom open; GateScanProcessed → Reverb → Guard UI

4.0 AI Parking Detection

AI POST /api/ai-parking/occupancy + plate-lookup → D4 parking_slots, cache
MJPEG stream proxied: /admin|guard/ai-parking/stream → :8090

5.0 Violation Management

AI events or Guard POST /guard/violations → D5 violations_log + evidence
ViolationEnforcementService → strikes, suspensions, email alerts

6.0 Reports & Administration

All hardware/AI APIs use token auth (X-RFID-TOKEN, X-AI-TOKEN). MongoDB holds 21 domain collections.
GET /admin/reports, export-pdf, export-excel → gate_logs, violations, parking
Admin: RFID, users, parking slots, settings, live cameras